

Junior Arroyo

PHD CANDIDATE

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Education

Johns Hopkins University

Baltimore, MD, United States

PH.D. IN BIOMEDICAL ENGINEERING

Aug 2021

- Research work: Laser safety, photoacoustic imaging, numerical modeling.
- Advisor: Prof. Muyinatu A. Lediju Bell

Pontifical Catholic University of Peru

Lima, Peru

M.SC. IN DIGITAL SIGNAL AND IMAGE PROCESSING

Mar 2015 - Dec 2016

- Thesis: "Evaluation of Elastographic Techniques Generated by Means of External Vibration."
- Advisors: Prof. Benjamin Castañeda (Pontifical Catholic University of Peru) and Prof. Tim Salcudean (University of British Columbia)

Pontifical Catholic University of Peru

Lima, Peru

B.SC. IN PHYSICS

Mar 2009 - Dec 2014

- Thesis: "Generation of Interference Patterns Using Quantitative Sonoelastography and Holography"
- Advisor: Prof. Benjamin Castañeda (Pontifical Catholic University of Peru).

Research experience

PULSE Lab - Johns Hopkins University

Baltimore, MD, United States

GRADUATE STUDENT

Aug 2021

- **Project: Computational and Experimental Approach to Determine Laser Safety of Liver, Ureter, and Skin Tissue.**
- Supervisor: Muyinatu A. Lediju Bell
- Research areas: laser safety, photoacoustic imaging, computational modeling, noninvasive and minimally invasive surgical guidance.

Polymers and Composite Materials Laboratory - Pontifical Catholic University of Peru

Lima, Peru

RESEARCH ASSISTANT

Dec 2017 - May 2019

- **Project: Use and Transformation of Agroforestry Waste for the Production of Ecological Plastics.**
- Supervisor: Fernando G. Torres.
- Research areas: Tailoring of nanoparticles, structure-property analysis of biological structures (seashells), molecular dynamics of biopolymers, multiparametric characterization of nanocomposites: morphology, conductivity, mechanical, and thermal properties.
- Additional task: Constant feedback and guidance to undergraduate members of the laboratory on their research work.

Macromolecular Engineering Group - Department of Chemistry and Properties of Polymeric Materials - Institute of Science and Technology of Polymers (ICTP-CSIC)

Madrid, Spain

GUEST RESEARCHER

Feb 2018

- Supervisor: Daniel López.
- Research areas: Morphological, dielectric, and structural characterization of native starch and starch-based nanoparticles.

Medical Imaging Laboratory - Pontifical Catholic University of Peru

Lima, Peru

RESEARCH ASSISTANT

Mar 2015 - Dec 2016

- **Project: Interinstitutional Research Applied to Improved Detection and Diagnosis of Tuberculosis and Gynecological Cancer.**
- Supervisor: Benjamin Castañeda, Roberto Lavarello.
- Research areas: Ultrasound imaging, elastography, optimization, simulation of waves propagation.

Robotics and Control Laboratory - University of British Columbia

British Columbia, Canada

GUEST RESEARCHER

Jan 2016

- Supervisor: Tim Salcudean.
- Research areas: Vibro-elastography, quantitative sonoelastography, ex-vivo tissue handling.

Medical Imaging Laboratory - Pontifical Catholic University of Peru

Lima, Peru

UNDERGRADUATE RESEARCH ASSISTANT

Jun 2014 - Dec 2014

- **Project: Application of Quantitative Sonoelastography as a Tool for Breast Cancer Detection.**
- Supervisor: Benjamin Castañeda.
- Research areas: Elastography, ultrasound imaging, quantitative sonoelastography.

Publications

JOURNALS

- **J. Arroyo**, J. Zhang, N. Venkatayogi, M. Kakkar, J. Zhang, A. Maxwell, K. Gabrielson, and M.A. Lediju Bell (2025). "In Vivo Photoacoustic Imaging of Swine Ureters Injected with Methylene Blue". *Journal of Biomedical Optics* (Accepted with minor corrections)
- **J. Arroyo**, J. Zhang, and M.A. Lediju Bell (2025). "Motion-based dynamic light delivery to minimize laser-related thermal damage while preserving photoacoustic image quality". *Journal of Biomedical Optics* 30(5), 056008-056008.
- J. Zhang, **J. Arroyo**, and M.A. Lediju Bell (2025). "Multispectral photoacoustic imaging of breast cancer tissue with histopathology validation". *Biomedical Optics Express* 16(3), 995-1005.
- **J. Arroyo**, A. Sharma, J. Zhang, and M.A. Lediju Bell (2024). "Predictive model for laser-induced tissue necrosis with immunohistochemistry validation". *Biophotonics Discovery* 1(2), 025003-025003.
- **J. Arroyo**, T.J. Marini, A.C. Saavedra, M. Toscano, T.M. Baran, K. Drennan, and B. Castaneda, (2022). "No sonographer, no radiologist: New system for automatic prenatal detection of fetal biometry, fetal presentation, and placental location". *PloS one* 17(2), e0262107.
- M. Toscano, **J. Arroyo**, A.C. Saavedra, T.J. Marini, T.M. Baran, K. Drennan, and B. Castaneda (2022). "Testing a deep learning algorithm for automatic detection of prenatal ultrasound for under-resourced communities". *American Journal of Obstetrics Gynecology* 226(1), S353-S354.
- O.P Troncoso, F.G. Torres, **J. Arroyo**, K.N. Gonzales, M. Fernández-García, and D. López (2020). "Mechanical properties of calcite-and aragonite-based structures by nanoindentation tests". *Bioinspired, Biomimetic and Nanobiomaterials* 9(2), 112-121.
- F.G. Torres, J.P. Mayorga, C. Vilca, **J. Arroyo**, P. Castro, and L. Rodriguez (2019). "Preparation and characterization of a novel starch-chestnut husk biocomposite". *SN Applied Sciences* 10, pp.1158.
- F.G. Torres, **J. Arroyo**, R. Alvarez, S. Rodriguez, and O.P. Troncoso (2019). "Carboxymethyl κ/ι -hybrid carrageenan doped with NH_4I as a template for solid bio-electrolytes development". *Materials Chemistry and Physics* 223, pp.659-665.
- F. G. Torres, **J. Arroyo**, C. Tineo, O.P. Troncoso (2019). "Tailoring the properties of native Andean potato starch nanoparticles using acid and alkaline treatments". *Starch-Stärke* 71(3-4), p.1800234.
- F.G. Torres, **J. Arroyo**, and O.P. Troncoso (2019). "Bacterial cellulose nanocomposites: An all-nano type of material". *Materials Science and Engineering: C* 98, pp.1277-1293.
- F.G. Torres, R. Ccorahua, **J. Arroyo**, and O.P. Troncoso (2019). "Enhanced conductivity of bacterial cellulose films reinforced with NH_4I -doped graphene oxide". *Polymer-Plastics Technology and Materials*, pp.1-11.
- F.G. Torres*, **J. Arroyo**, R. Alvarez, S. Rodriguez, O.P. Troncoso, and D.López (2018). "Molecular dynamics of carboxymethyl κ/ι -hybrid carrageenan films doped with NH_4I ". *Polymer-Plastics Technology and Engineering* 58, pp.889-902.

CONFERENCES

- **J. Arroyo**, J. Zhang, and M.A. Lediju Bell (2025). "A unified computational and experimental approach to determine laser safety of liver tissue". *Optical Interactions with Tissue and Cells XXXVI*, 13317, pp. 19-26.

- J. Zhang, **J. Arroyo**, and M.A. Lediju Bell (2024). "Multispectral photoacoustic imaging of breast cancer tissue". *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, pp. 1-4.
- **J. Arroyo**, M.T. Graham, E.A. Gonzalez, L. Zhao, M.A.L. Bell (2022, October). "Effect of light source motion on photoacoustic image quality and safety". In *Laser Science*, pp. JW5B-44, Optica Publishing Group.
- **J. Arroyo**, A.C. Saavedra, L. Tamayo, M. Egoavil, B. Ramos, and B. Castaneda (2021, February). "Automatic fetal presentation diagnosis from ultrasound images for rural zones: head location as an indicator for fetal presentation". In *Medical Imaging 2021: Computer-Aided Diagnosis* 11597, pp. 281-289
- A.C. Saavedra, **J. Arroyo**, L. Tamayo, M. Egoavil, B. Ramos, and B. Castaneda (2020, September). "Automatic ultrasound assessment of placenta previa during the third trimester for rural areas". In *2020 IEEE International Ultrasonics Symposium (IUS)*, pp. 1-4.
- **J. Arroyo**, A.C. Saavedra, J. Guerrero, P. Montenegro, J. Aguilar, J.A. Pinto, J. Lobo, T. Salcudean, R. Lavarello, and B. Castañeda (2018). "Breast elastography: classification of benign and malignant cancer based on absolute elastic modulus measurement using Vibro-elastography". *SPIE Medical Imaging 2018: Image Perception, Observer Performance, and Technology Assessment*. Vol.10577. International Society for Optics and Photonics, pp.105771E.
- A.C. Saavedra, **J. Arroyo**, F. Zvietcovich, R. Lavarello, and B. Castaneda (2018). "In vivo estimation of the Young's modulus in normal human dermis". *Engineering in Medicine and Biology Society (EMBC), 2018 40th Annual International Conference of the IEEE. IEEE*, pp.3456-3459
- A.C. Saavedra, A. Coila, **J. Arroyo**, B. Castaneda, and R. Lavarello (2018). "In vivo Attenuation Coefficient Estimation in the Healthy Forearm and Thigh Human Dermis". *IEEE International Ultrasonics Symposium (IUS) 2018. IEEE*, pp. 1-4.
- **J. Arroyo**, and B. Castaneda (2017). "Shear wave estimation by using Shear Wave Holography with normal vibration: Preliminary results". *Engineering in Medicine and Biology Society (EMBC), 2017 39th Annual International Conference of the IEEE. IEEE*, pp.3004-3007

Honors & Awards

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|------|--|-------------------|
| 2008 | Fellowship "PUCP - Fe y Alegria" , Full-tuition scholarship for undergraduate studies. | <i>Lima, Peru</i> |
| 2014 | Research Initiation Support Program , Research topic: "Generation of interference patterns using quantitative sonoelastography and holography." | <i>Lima, Peru</i> |

Teaching experience

Johns Hopkins University

TEACHING ASSISTANT, INTRODUCTION TO MEDICAL IMAGING

Aug 2024

- Computer Tomography, Ultrasound, Magnetic Resonance Imaging

Pontifical Catholic University of Peru

TEACHING ASSISTANT, INTRODUCTION TO PHYSICS

Mar 2015 - Dic 2015

- Kinematics and dynamics.

TEACHING ASSISTANT, PHYSICS 1

Mar 2015 - Dic 2017

- Kinematics and vector dynamics.

TEACHING ASSISTANT, PHYSICS 2

Mar 2015 - Jul 2015

- Elasticity, waves, fluids, and thermodynamics.

TEACHING ASSISTANT, DIGITAL SIGNAL AND IMAGE PROCESSING

Mar 2015 - July 2021

- Z-transform, Fourier transform, filter design, spatial filtering, image enhancement, segmentation.

References

Professor Muyinatu A. Lediju Bell, advisor

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JOHNS HOPKINS UNIVERSITY

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Professor Fernando G. Torres, employer

MECHANICAL ENGINEERING DEPARTMENT, POLYMERS AND COMPOSITE MATERIALS
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Professor Benjamin Castañeda, advisor

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